Telecommunication and IT Fundamentals

Course objectives:

This training is your gateway into the world of Telecommunication and IT. It is designed as the allrounder, helping you finding your way around modern telecommunication and IT concepts, terminology and technologies.

You will learn about the building blocks, architecture and operation of telecommunication and computer networks, enabling you to understand the different wired and wireless network standards on hand.
We'll have a glimpse on what makes the Internet tick and how to use it to offer an abundance of services, such as multimedia services, cloud services and many more in a controlled and secure environment.
Finally, the fast evolution of Mobile Networks warrants a closer look at defining architecture, concepts and services, shaping the telecommunication world every day.
Advancing into future technologies, a module about quantum computing and quantum communication rounds up this training course.

After attending this course, you will be able to participate in discussions and make your business decisions with more confidence.

Duration and Instructional methods:

Option 1: On-Site Training: 3 days,
Classroom-style lecture using Power-point slides with interactive discussion and Questions & Answers after each module.

Option 2: Virtual Classroom: 3 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 6 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to participants from the technology, marketing and management field requiring an overview about the different telecommunication and IT solutions, fixed and wireless network standards, architectures and implementations.

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Course Contents:

Basic Concepts, Terminology and Telephony Networks

- Telecommunication History
- Block Diagram of Basic Communication System
- Standardization Bodies in Telecommunication
- The Electromagnetic Signal
- Media in Telecoms: Audio and Video signals, Textual information
- Modulation / Multiplexing / Switching
- Telephone Networks: PSTN, ISDN and Private Networks (PABX)
- Numbering and Addressing in Telephone Networks
- Digital communication systems: A/D Conversion
- Carrier (Transmission) systems:
    - Plesiochronous Digital Hierarchy (PDH) systems
    - SDH/SONET systems
    - DWDM and optical transmission networks (OTN)
- The OSI Reference Model
- Signaling Technologies and Principles:
    - Signalling System No.7 (SS7)
    - SIP Signaling and DIAMETER

Data Communication and Data Networks

- Wired and Wireless Transmission Media:
    - Twisted Pair / Coaxial / Fiber Optical Cables
    - Microwave Radio / Satellite / Infrared / Laser Communication Systems
- Data Networking Types and Classification
- LAN Architecture - Ethernet
- MAN / WAN basics:
    - xDSL Technology and variants
    - Cable Modem Networks (DOCSIS)
    - Fiber Optical Networks and FTTx
- Backbone Networks and Technologies:
    - MPLS
    - Metro Ethernet
- The Internet
- TCP/IP Networks and Applications
- Security Aspects for TCP/IP networks
- VoIP, IPTV and Streaming Services

- NFV and SDN
- Cloud Services – Role of Data Centers
- Internet of Things (IoT)and M2M
- Overview Artificial Intelligence, Machine Learning and Big Data
- Quantum Computing and Quantum Communication Introduction

Wireless Systems and Mobile Networks

  - WiFi and WiGig
  - WiMAX
  - Bluetooth, NFC, Li-Fi
- Mobile Networks / Cellular Networks
- GSM / GPRS / EDGE
- 3G UMTS and HSPA
- LTE / LTE Adv. / LTE Adv. Pro
- LTE for Public Safety and Mission Critical Applications
- 5G Networks and outlook towards 6G and 7G
- Het-Net concept and Femto-Cells
- Services: Quadruple-Play
- The IP Multimedia System (IMS)
- Voice / Video Services over data networks: VoLTE / VoWiFi /ViLTE, Vo5G, VoNR
- Mobile Backhaul Solutions and Cloud-Based Mobile Networks Architecture
- Low-Power Wide-Area Networks (LPWAN):
  - LoRaWAN, Sigfox, NB-IoT

Pre-requisites:

Interest in Telecommunications and IT.

Course Code: Basico1